

FORT NELSON, BC, Canada

WMO: 719450

Lat: 58.8415N

Lon: 122.5741W

Elev: 1253

StdP: 14.04

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-27.1	-41.1	0.5	-33.3	-35.0	0.8	-26.9	16.1	15.6	13.3	8.9	1.1	210	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	82.3	62.3	79.1	61.1	76.0	59.7	65.0	77.2	63.2	74.6	61.5	72.3	6.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	83.2	68.7	58.9	77.8	66.7	57.2	73.1	65.2	30.6	77.0	29.2	74.8	28.0	72.4	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.5	13.4	11.7		-38.0	88.1	6.5	2.7	-42.7	90.0	-46.4	91.6	-50.1	93.1	-54.8	95.1	
				-38.2	68.3	6.4	2.4	-42.8	70.0	-46.5	71.4	-50.1	72.8	-54.8	74.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-3.3	6.0	17.3	37.0	50.1	59.0	63.1	59.5	49.1	33.9	10.7	-1.3
	DBStd	25.93	14.63	12.68	14.54	9.48	8.24	5.94	5.17	6.29	7.07	8.39	11.79	12.65
	HDD50	7772	1651	1232	1015	393	98	7	1	7	99	502	1178	1589
	HDD65	12155	2116	1652	1480	839	464	193	97	190	478	964	1628	2054
	CDD50	1162	0	0	0	4	101	277	406	300	71	3	0	0
	CDD65	72	0	0	0	0	2	13	38	19	0	0	0	0
	CDH74	1099	0	0	0	2	71	253	477	274	21	1	0	0
	CDH80	200	0	0	0	0	8	44	98	49	1	0	0	0
Wind	WSAvg	4.1	2.7	3.3	4.5	5.3	5.5	5.1	4.7	4.3	4.3	3.8	3.0	2.6
Precipitation	PrecAvg	18.5	0.9	0.7	0.8	0.8	1.9	2.8	3.3	2.8	1.7	1.3	1.0	0.8
	PrecMax	29.6	3.0	1.6	2.5	2.8	5.1	7.4	8.0	8.1	4.8	4.5	3.2	2.0
	PrecMin	12.6	0.2	0.2	0.1	0.0	0.3	0.8	0.7	0.5	0.0	0.0	0.1	0.0
	PrecStd	3.5	0.6	0.4	0.5	0.7	1.1	1.5	1.5	1.7	1.1	0.8	0.6	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	40.5	54.9	68.6	80.9	84.5	86.9	85.1	77.3	63.8	43.2	37.8
		MCWB	34.1	33.8	41.8	49.8	56.9	61.8	65.2	65.0	60.2	50.2	35.1	31.9
	2%	DB	28.9	33.9	48.7	62.0	75.5	80.4	82.7	80.6	70.8	56.2	35.5	28.1
		MCWB	26.4	28.9	37.9	46.1	55.1	60.3	63.3	63.0	57.0	44.8	31.1	25.7
	5%	DB	21.1	28.6	43.7	57.6	71.4	76.6	79.3	76.9	66.5	50.6	30.3	21.9
		MCWB	19.9	26.0	34.7	43.1	53.1	58.4	62.2	61.3	53.5	42.2	28.2	20.9
	10%	DB	15.3	24.2	38.7	53.6	67.3	73.0	75.9	73.2	62.8	46.0	26.1	15.5
		MCWB	14.4	22.2	31.6	40.8	51.4	57.1	60.7	59.8	51.9	40.0	24.7	14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.8	34.6	42.1	50.5	59.9	64.3	68.1	67.5	61.7	51.4	35.9	32.7
		MCDB	39.0	40.3	54.0	67.2	75.0	78.1	80.6	79.6	73.7	60.8	41.8	37.0
	2%	WB	26.4	29.3	38.1	46.8	56.7	61.9	65.5	64.8	58.4	46.1	31.4	25.8
		MCDB	28.8	33.5	48.0	60.7	71.8	74.9	78.1	76.6	68.0	54.4	34.8	27.8
	5%	WB	20.0	25.9	35.1	43.9	54.4	60.3	63.7	62.8	55.3	42.8	28.2	20.9
		MCDB	21.1	28.6	43.2	55.9	68.4	72.6	75.3	73.9	63.5	49.2	30.3	22.0
	10%	WB	14.5	22.2	31.9	41.5	52.0	58.6	62.1	60.9	52.9	40.1	24.7	14.6
		MCDB	15.3	24.2	38.4	52.4	65.4	70.2	72.6	70.9	60.6	45.8	26.1	15.5
Mean Daily Temperature Range	5% DB	MDBR	13.1	18.5	21.6	21.6	23.1	21.3	20.4	21.1	20.7	14.8	12.3	12.1
		MCDBR	20.2	24.5	26.8	29.9	31.3	28.9	27.2	27.8	28.0	24.9	18.0	18.5
		MCWBR	18.0	20.1	17.6	16.4	14.9	12.5	11.6	13.0	15.2	15.3	14.2	16.1
	5% WB	MCDBR	20.2	23.7	26.5	28.4	28.2	25.1	24.2	25.3	26.4	23.6	17.3	18.2
		MCWBR	17.9	19.5	17.6	16.2	13.7	11.6	11.2	12.8	15.4	15.1	13.9	15.9
Clear-Sky Solar Irradiance	taub	0.218	0.246	0.274	0.328	0.349	0.360	0.365	0.351	0.311	0.291	0.230	0.196	
	taud	2.240	2.325	2.306	2.352	2.422	2.418	2.421	2.475	2.589	2.552	2.326	2.126	
	Ebn at Noon	207	257	279	279	280	278	273	269	267	239	201	171	
	Edh at Noon	13	20	28	33	33	33	33	29	22	17	12	10	
All-Sky Solar Radiation	RadAvg	158	435	921	1473	1709	1762	1702	1413	933	462	188	95	
	RadStd	11	34	57	62	125	124	131	125	73	42	22	6	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.88	+3.64	+4.75	N/A	N/A	N/A	N/A	-259	N/A	N/A
Regional (2 neighbors)		+0.95	+3.78	+5.42	N/A	N/A	N/A	N/A	-293	N/A	N/A

Nomenclature: See separate page